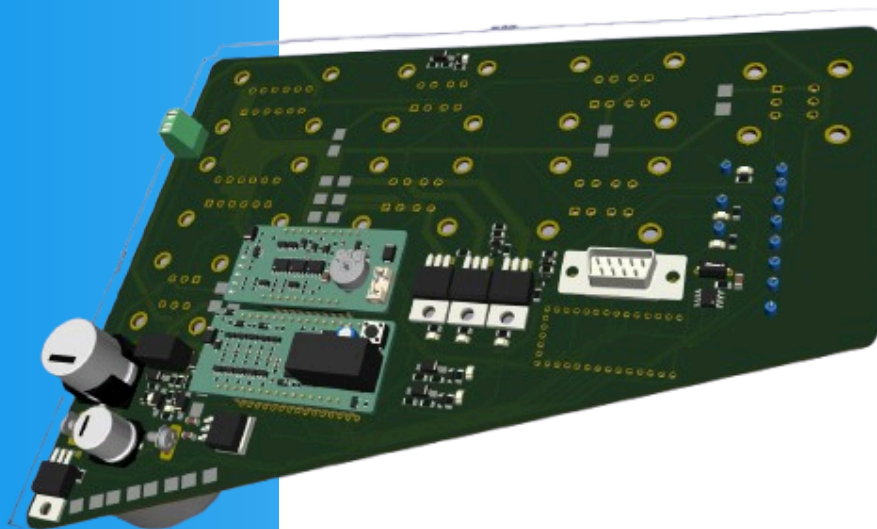




ENGG3200 Report



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2025



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Introduction

This report documents the design, manufacture, and commissioning of the 2025 iteration of the NU Racing MOTHER board (V2). This project builds on the work of the 2024 MOTHER board (V1.2), which was designed by Jacob Lukes. The goal for 2025 was to address serviceability limitations to simplify diagnostics, and improve system reliability in regards to noise and potential field failures.

Key differences from the previous year's board include the integration of an interpose relay to strengthen shutdown signal integrity, upgrades to MOSFETs for improved thermal and switching performance, and refined fuse specifications. The PCB layout was also optimized for installation on the opposite side of the HIP, creating easier access for servicing and testing.

Throughout the design and development process, Jacob Lukes provided invaluable mentorship and technical insight. His documentation, previous schematic designs, and direct feedback on layout reviews were critical to ensuring a smooth transition between board versions and successful implementation of updated features.

Initial Design

Background

The NU23 MOTHER board, developed by Jacob Lukes, combined the functionality of the Central Electric Node (CEN) and Rear Electronic Node (REN) into a single PCB. While it significantly reduced system complexity, the board still faced challenges including difficult service access, overheating MOSFETs, and inconsistent shutdown circuit behavior. The 2025 revision aimed to solve these limitations by refining the PCB design, re-specifying critical components, and improving system modularity.

The MOTHER board V2 was redesigned to live on the opposite side of the chassis from the HIP, allowing better cable routing and service access. Many of the original REN functions such as cooling control, brake light switching, and the sounder circuit remained but were optimized with new MOSFETs and smarter layout.

Board Evolution Figures

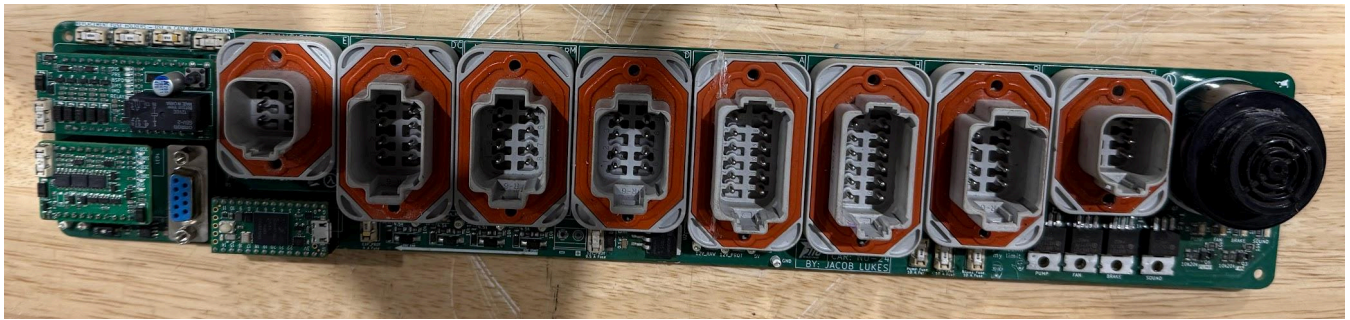


Figure 1: 2024 MOTHER Board Front View

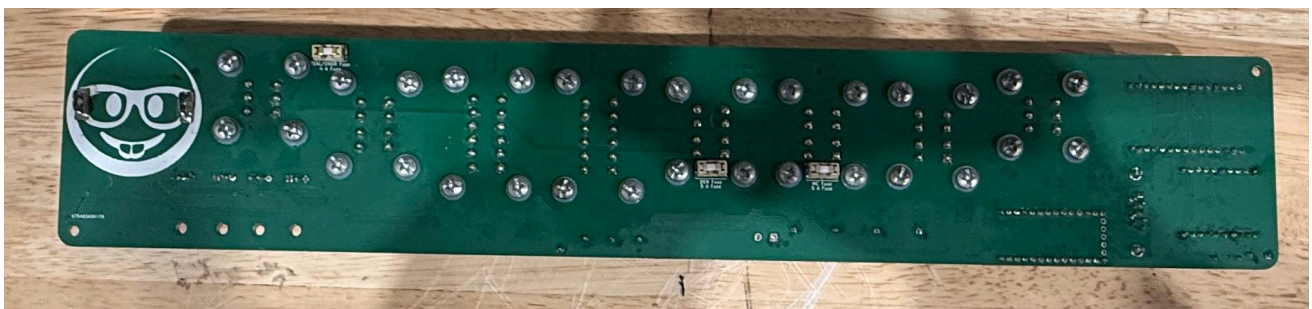


Figure 2: 2024 MOTHER Board Rear View

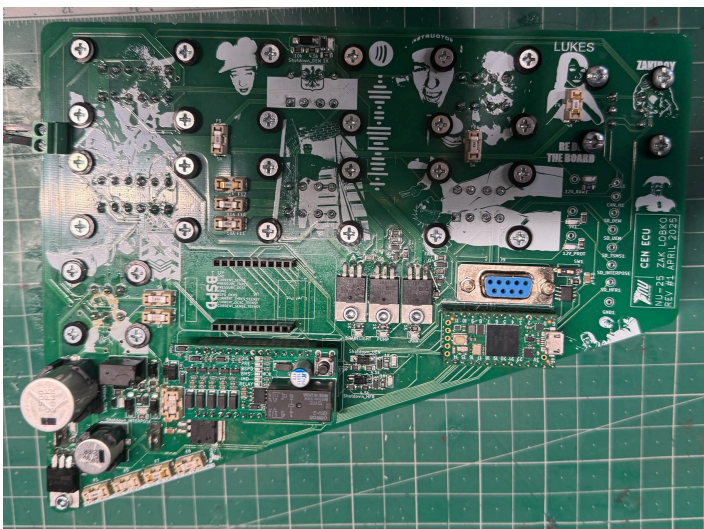


Figure 3: 2025 MOTHER Board Front View

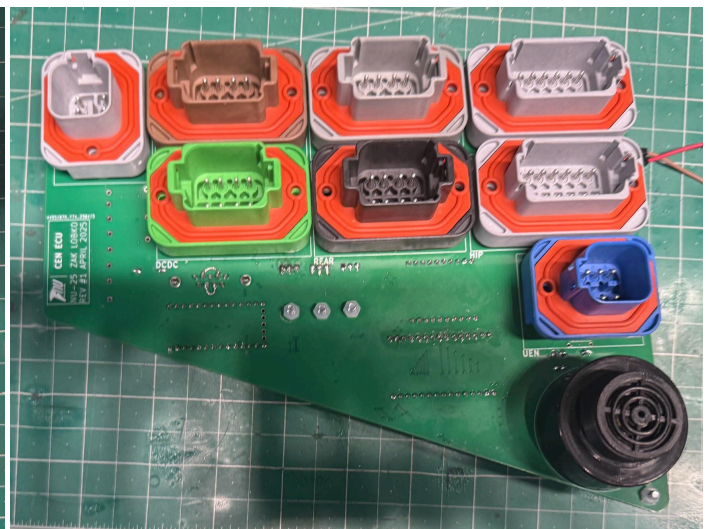


Figure 4: 2025 MOTHER Board Rear View

Scope

The 2025 iteration of the MOTHER board was developed as part of the NU Racing LV system architecture update. The primary objective of this project was to enhance the serviceability, performance, and safety compliance of the Central Electric Node (CEN) system by implementing hardware and layout changes informed by testing and feedback from the 2024 V1.2 version.

This year's scope focused on executing schematic and physical design improvements, revising component selections, and validating performance through system-level integration tests with the BSPD, HIP, and LVD boards. The updated design also had to accommodate physical constraints within the vehicle by relocating the board to the opposite side of the HIP.

Key components of the scope included:

- Redesigning the schematic to integrate component substitutions and interpose circuitry.
- Modifying the layout to improve routing, fuse access, and modularity.
- Validating new MOSFETs and connector configurations in real-world conditions.
- Ensuring FSAE rule compliance with respect to shutdown behavior, current sense analog sensor, and fault protection.
- Collaborating closely with BSPD and HIP designers for seamless integration.
- Producing and populating the final board for use in NU24 track days.

Design Changes from MOTHER Board V1.2

The 2025 MOTHER board features several critical changes in both its electrical and mechanical design compared to the 2024 V1.2 version. These changes were implemented to improve reliability, simplify manufacturing, and meet new packaging requirements driven by updates to the vehicle layout.

One of the most significant design changes was the upgrade of the low-side switching MOSFETs. The IRLB8721 devices used previously were replaced with IRLB3813 MOSFETs, which offer better thermal performance and increased current handling. Similarly, the smaller gate-driving transistors were changed from 2N7002 to DMN6075S-7, providing improved switching characteristics and reduced failure risk during soldering or operation.

Connector types were updated throughout the board. The original 8-pin DT connectors were replaced with keyed AT connectors. These connectors are easier to install correctly and help prevent reverse or incorrect mating, addressing a common point of failure during integration and servicing.

Fuse values were reassessed and re-specified based on the measured performance of components in the 2024 vehicle. The brake light and radiator fan, previously fused at 10A, were each lowered to more appropriate values aligned with their actual current draw. This change improves protection and prevents unnecessary risk of overheating.

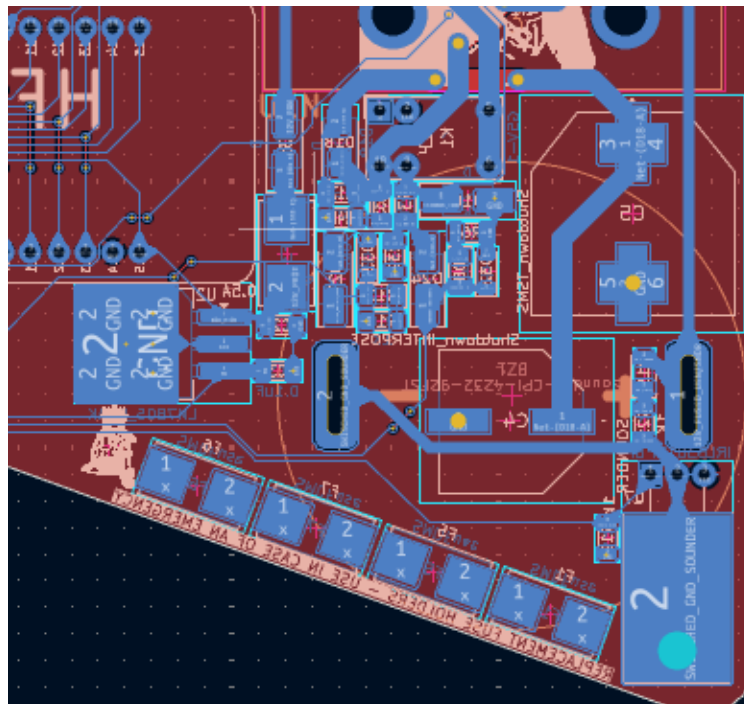


Figure 5: SHUTDOWN_INTERPOSE relay integration.

A key new addition to the design was the inclusion of an interpose relay circuit. This relay boosts the shutdown signal voltage before it reaches the accumulator, ensuring reliable HV enablement even under marginal conditions. The new signal SHUTDOWN_INTERPOSE was clearly marked on the schematic and PCB layout to reflect this new control logic.

Mechanically, the board was reshaped to fit opposite the HIP on the chassis, simplifying wiring and improving accessibility for diagnostics. The TSAL connector was also renamed to UEN for consistency with the system architecture. A new footprint for the BSPD was added based on the specification provided by Jayson Mills, allowing for integration with BSPD V3.

These changes reflect an iterative improvement of the MOTHER board that builds on the strengths of the 2024 version while addressing its limitations in both design and usability.

Schematic Overview

The 2025 schematic includes several upgrades over the 2024 version. These improvements aim to increase circuit reliability and resilience, while also streamlining debugging and integration processes. The MOSFETs used for low-side switching circuits, including those for the sounder, brake light, cooling pumps, and radiator fan, were changed from IRLB8721 to IRLB3813. These new devices provide enhanced thermal and current performance. The gate driver transistors were also updated from 2N7002 to DMN6075S-7 to ensure more consistent operation under varying signal conditions.

A notable addition to the shutdown circuit is the interpose relay, which was introduced to boost the 12V signal before it reaches the accumulator. This change, along with the clearly defined SHUTDOWN_INTERPOSE signal, provides more robust system control and diagnostics. Input protection was expanded by standardizing the use of Zener diodes and voltage dividers across all monitored lines, each supported by individual test points to aid commissioning.

Breakout board integration for BSPD and HFR has been made more modular, allowing easier isolation of issues during debugging. The I/O section was cleaned up and re-labeled according to subsystems, with all connectors now using keyed AT designs instead of DT to prevent incorrect connections. A centralized ground test block was also added to simplify multimeter use when diagnosing faults.

The schematic was organized by function, dated, and clearly annotated. Net labels, voltage references, and silkscreen callouts are used consistently to improve readability and traceability across the board.

PCB Layout and Physical Changes



Figure 6: 2024 board on scales

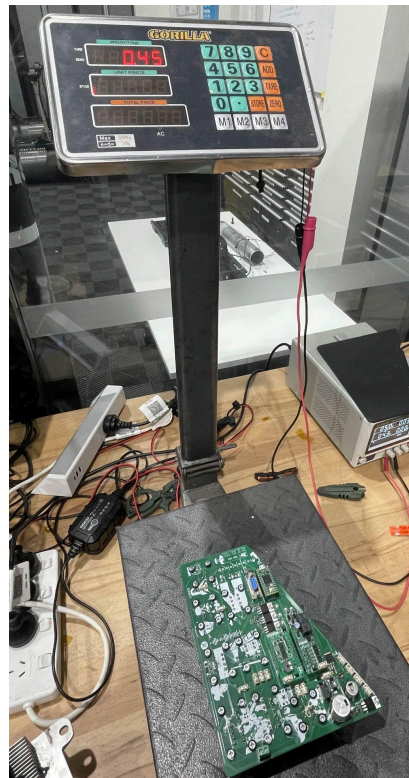


Figure 7: 2025 board on scales.

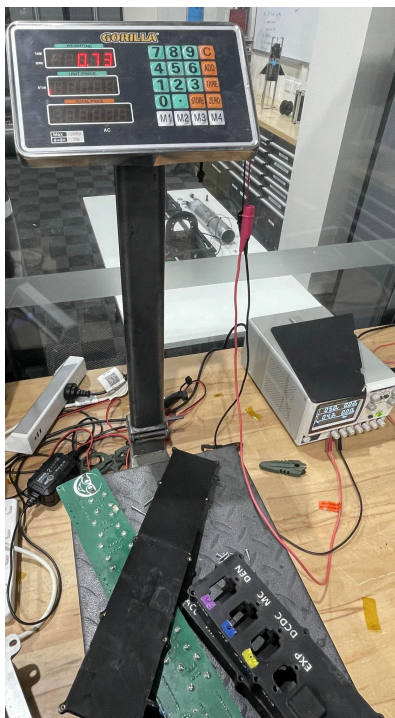


Figure 8: 2024 board with case

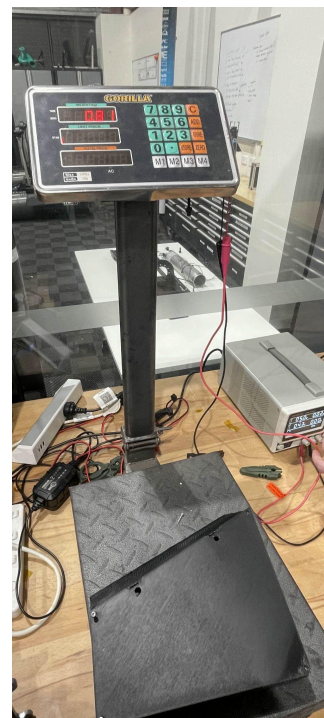


Figure 9: 2025 board on with case.

After the development process, both the 2024 and 2025 MOTHER boards were weighed for comparative analysis. With their external student-designed enclosures, the 2024 board came in at 730 grams, while the updated 2025 version weighed slightly more at 810 grams. When measured without their cases, the weight difference between the two designs was negligible 440 grams for the 2024 version and 450 grams for the 2025 board. This confirms that the additional functionality and layout changes introduced in the 2025 design did not result in any significant increase in base PCB weight, and that the majority of the added weight is attributable to the updated casing. This may be useful for future teams considering packaging constraints in the vehicle.

The PCB layout for the 2025 MOTHER board was redesigned with both performance and serviceability in mind. One of the key changes was the relocation of the fuse block to a more central and accessible position, making trackside servicing quicker and less intrusive. Breakout board areas for the BSPD and HFR were enlarged and outlined, with sufficient clearance around the headers to allow easy removal and replacement.

Routing was revised throughout the board. High-current traces, particularly those connected to the radiator fan and cooling pumps, were widened and spaced for improved voltage stability and reduced heating. Signal lines were rerouted to avoid congestion, minimize noise coupling, and reduce unnecessary via usage. This not only improved circuit performance but also enhanced reliability under vibration and thermal cycling.

The area around the Teensy and CAN transceiver was cleaned up, reducing trace length and minimizing stub lines that previously introduced noise. The board's mechanical footprint was also reshaped to fit the new mounting location opposite the HIP.

To help with on-the-fly diagnostics and servicing, silkscreen improvements were made across the board. All test points are found to the left hand side of the board which are easily accessible while the board is in the car. In reference to Figure 10 below the top lid of the board enclosure can be removed making it easy to see all the shutdown and BSPD LEDs which reduces time spent on diagnosing problems.



Figure 10: 2025 in enclosure with lid off

Manufacturing and Assembly

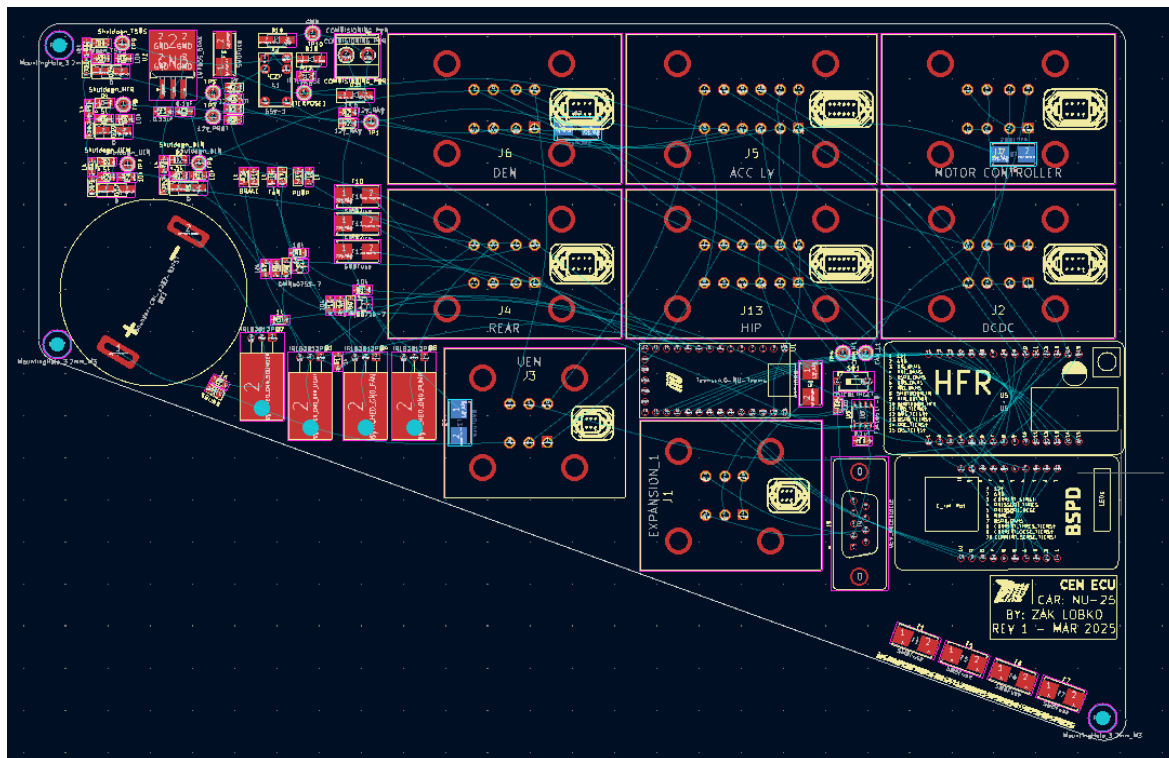


Figure 11: CEN V1

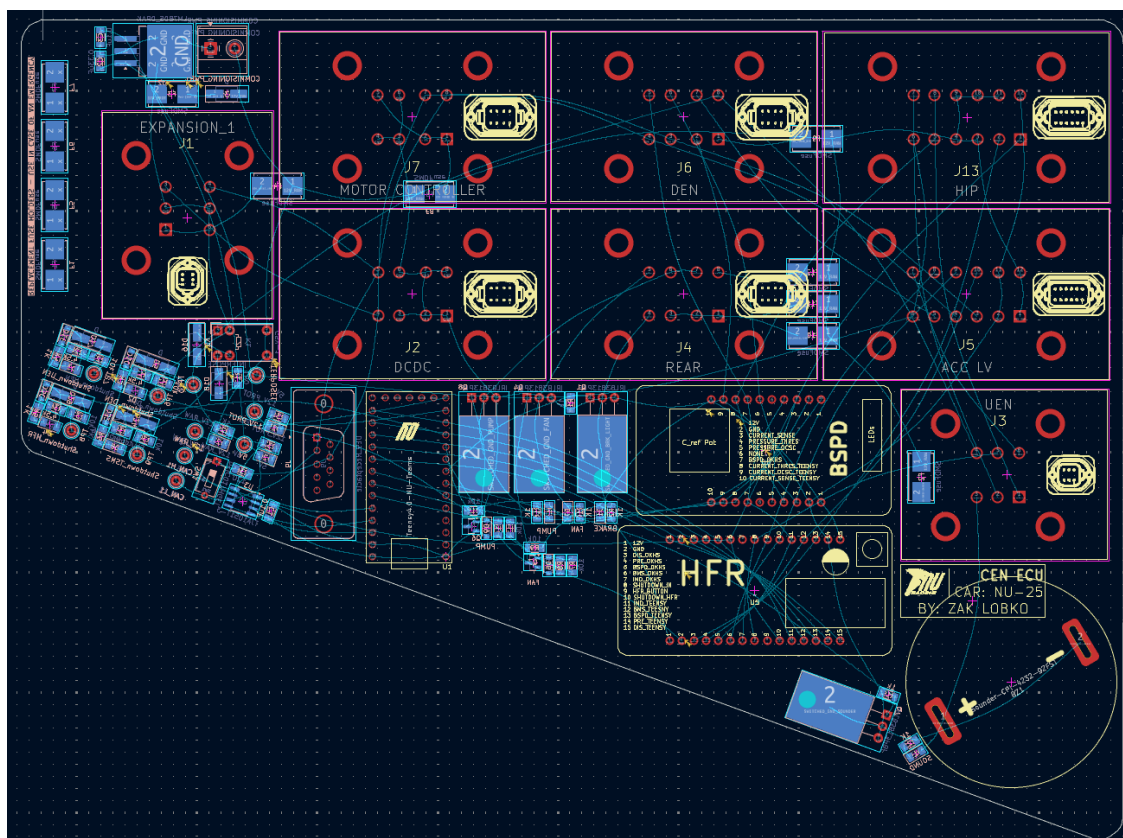


Figure 12: CEN V2

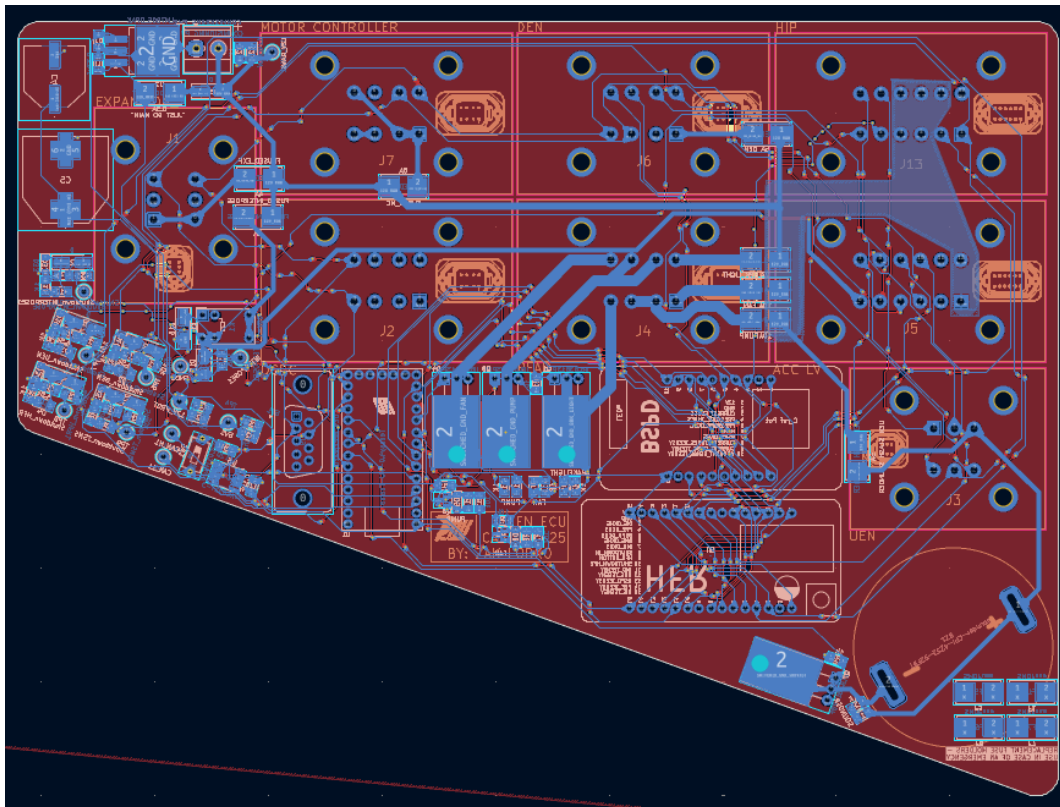


Figure 12: CEN V2.1 (Final iteration)

The manufacturing and assembly phase of the 2025 MOTHER board went through three major layout iterations before the final design was settled. These iterations reflected progressive refinements in trace routing, breakout board placement, and connector alignment. Early versions had less optimized fuse placement and more congested routing near the Teensy and CAN transceiver. Through regular reviews and feedback, from Jacob Lukes, the layout evolved to improve manufacturability, structural integrity, and service access.

The final PCB was manufactured by JLCPCB and assembled in the MCHA lab using a solder paste stencil and reflow process for SMD components. Hand soldering was used for through-hole parts, with particular care taken for sensitive MOSFETs and the keyed AT connectors. These AT connectors were soldered post-reflow due to their larger physical profile and mechanical stress characteristics.

Each subcircuit was tested individually as it was soldered onto the board. This approach helped isolate and resolve faults before full integration, reducing rework and commissioning time. A multimeter was used extensively throughout to verify continuity, signal lines, and input protection paths.

This iteration also benefited from visual and structural silkscreen elements which improved identification during population and made manual inspection faster and more effective.

Commissioning and Testing

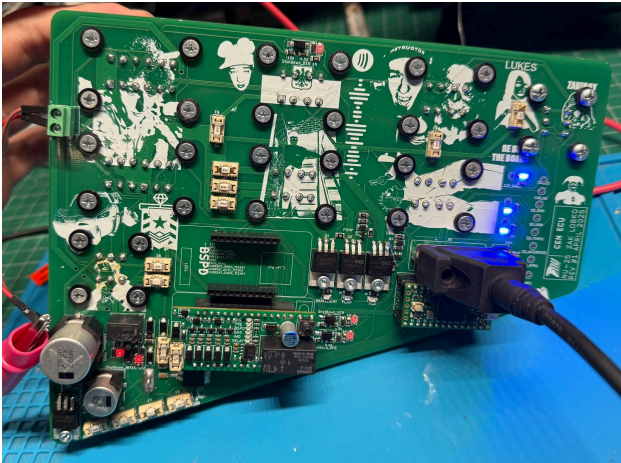


Figure 13: CAN Communication Check

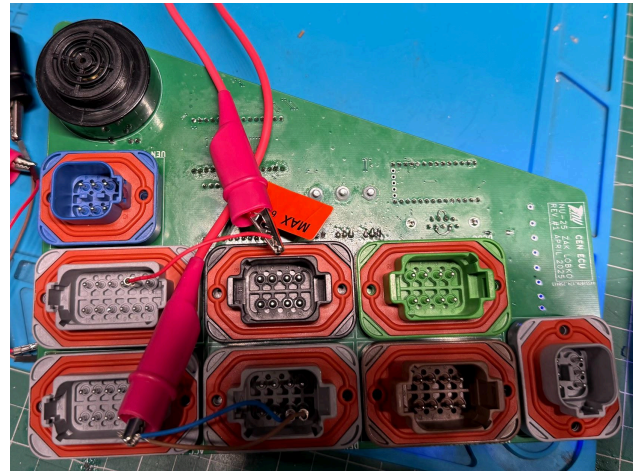


Figure 14: Verify Shutdown Circuits

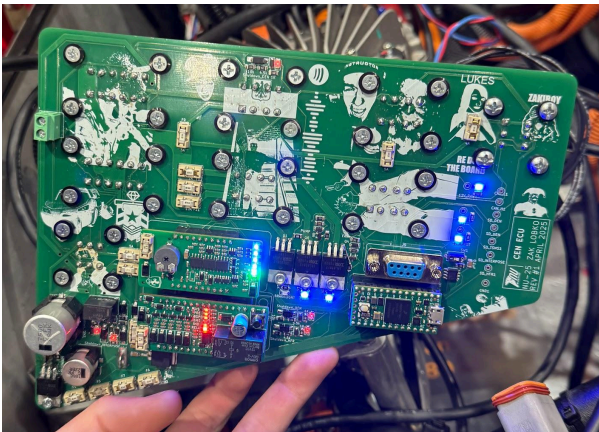


Figure 15: BSPD & HFR integration test

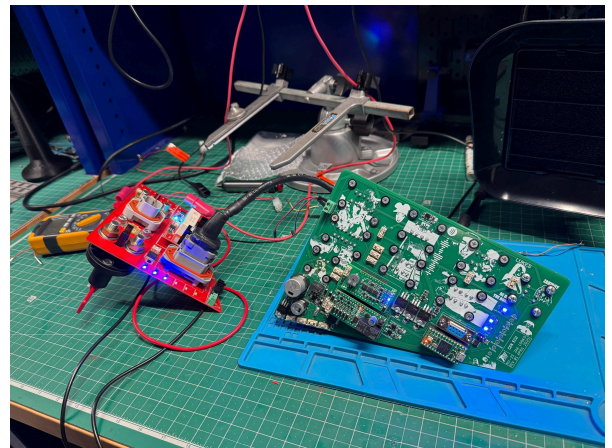


Figure 16: HIP integration test

Commissioning of the 2025 MOTHER board followed a structured process guided by a unit test checklist. Each subsystem, cooling, sounder, brake light, shutdown circuits, and CAN, was tested in isolation before moving to full-board power-up. The Teensy microcontroller was flashed and verified for CAN communication functionality early in the process to validate bus stability.

Multimeter testing was conducted on all fused and protected input paths, ensuring that the Zener and voltage divider networks correctly stepped 12V signals down to safe 3.3V logic levels. MOSFETs and their gate resistors were tested for correct low-side switching behavior in each application (cooling, pump, sounder, brake light).

Results from all commissioning steps were recorded, and photos were taken for key measurements and setups.

Integration and Field Testing

Following commissioning, the MOTHER board was integrated with the HIP, BSPD, and LVD systems and tested on-bench and in-vehicle. The first full system integration occurred on 25/05/2025 with all subsystems powered and communicating.

During initial integration, a mismatch between the Teensy firmware and the hardware pinout caused several failures in the shutdown and GND sensing pathways. This issue stemmed from uncoordinated pin reassignment and was corrected by updating the Teensy code.

An issue was observed where excessive capacitance remained on the shutdown line due to duplicated capacitor placement on both the MOTHER board and LVD. This caused the E-stop to take approximately 19 seconds to fully disengage. The capacitors were later removed from the LVD to resolve the issue.

Sounder malfunctions were also encountered. Specifically, after an E-stop event or initial start up, the system would sometimes fail to re-enable the sounder upon re-entering RTD. Quick power cycles (<5 seconds) did not restore function, but full power-downs did. This was suspected to be a hard fault latch within the microcontroller or associated driver circuitry.

Lessons Learned

The development and testing of the 2025 MOTHER board offered several key takeaways that will guide future designs and team practices. One of the most immediate observations was the fragility of the DMN6075S-7 gate driver MOSFETs during soldering. Their small package size made them prone to thermal damage during hand assembly. In future versions, more robust thermal handling procedures or alternative footprints may be needed.

The transition to keyed AT connectors had a substantial impact on integration efficiency and correctness. The design prevented misconnection errors that were previously observed with DT connectors, particularly in high-pressure install scenarios with junior students.

One of the most impactful design inclusions was the interpose relay. Its ability to ensure clean shutdown behavior under noisy or degraded conditions proved essential. It will remain a required feature in future MOTHER board versions.

The field test also highlighted some communication breakdowns between hardware and firmware development. Teensy pin reassignment was not mirrored in code updates, leading to initial test failures. This emphasizes the need for a unified hardware/software change tracking system.

Conclusions and Recommendations

The 2025 MOTHER board achieved its design and performance objectives, demonstrating marked improvements over the 2024 V1.2 version. By refining the electrical design, optimizing the PCB layout, and addressing component-level issues, the new board provided increased reliability and improved serviceability. The interpose relay proved essential in ensuring stable shutdown functionality under real-world conditions, while updated MOSFET selections and connector systems enhanced overall robustness and integration safety.

Testing confirmed that the board performed as expected across all subsystems. The shutdown circuit responded cleanly, CAN communication remained stable, and all outputs functioned reliably under load. While several integration issues were encountered, such as Teensy pin mapping and excessive capacitance on the shutdown line, these were resolved with minimal rework.